A FLAT FINITE MODULE OVER A POLYNOMIAL RING THAT IS FINITELY PRESENTED AT FINITELY MANY PRIMES IS FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Openness of flatness for a finite module over a Noetherian-free base, in the form that the flat locus is open. the author cites Algebra, Theorem algebra-theorem-openness-flatness at proof line 5813. Provides the localisation at which the auxiliary module becomes flat.
- A module generated by given elements on each piece of a finite cover is generated by them globally. the author cites Algebra, Lemma algebra-lemma-cover at proof line 5830. Concludes finite presentation from the local statement.
- Injectivity of the map from the base ring to the product of its localisations at the chosen primes, and flatness of the modules involved. used at proof lines 5818-5828. Makes the left vertical map of the displayed diagram injective.

Source reference. Original lemma 053A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6598–6598. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00RC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 998–998. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00EO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 542–542. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $S = R[x_1, \ldots, x_n]$ be a polynomial ring over R. Let M be an S-module. Assume*

(1) *there exist finitely many primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ of R such that the map $R \to \prod R_{\mathfrak{p}_j}$ is injective,*
(2) *M is a finite S-module,*
(3) *M flat over R, and*
(4) *for every prime $\mathfrak{p}$ of R the module $M_{\mathfrak{p}}$ is of finite presentation over $S_{\mathfrak{p}}$.*

Then M is of finite presentation over S.

Proof. Choose a presentation

$$0 \to K \to S^{\oplus r} \to M \to 0$$

of M as an S-module. Let $\mathfrak{q}$ be a prime ideal of S lying over a prime $\mathfrak{p}$ of R. By assumption there exist finitely many elements $k_1, \ldots, k_t \in K$ such that if we set $K' = \sum Sk_j \subset K$ then $K'_{\mathfrak{p}} = K_{\mathfrak{p}}$ and $K'_{\mathfrak{p}_j} = K_{\mathfrak{p}_j}$ for $j = 1, \ldots, m$. Setting $M' = S^{\oplus r}/K'$ we deduce that in particular $M'_{\mathfrak{q}} = M_{\mathfrak{q}}$. By openness of flatness, see Algebra, Theorem 00RC we conclude that there exists a $g \in S$, $g \notin \mathfrak{q}$ such that M'_g is flat over R. Thus $M'_g \to M_g$ is a surjective map of flat R-modules. Consider the commutative diagram

$$\begin{array}{ccc} M'_g & \longrightarrow & M_g \\ \downarrow & & \downarrow \\ \prod (M'_g)_{\mathfrak{p}_j} & \longrightarrow & \prod (M_g)_{\mathfrak{p}_j} \end{array}$$

The bottom arrow is an isomorphism by choice of $k_1, \ldots, k_t$. The left vertical arrow is an injective map as $R \to \prod R_{\mathfrak{p}_j}$ is injective and M'_g is flat over R. Hence the top horizontal arrow is injective, hence an isomorphism. This proves that M_g is of finite presentation over S_g. We conclude by applying Algebra, Lemma 00EO. $\square$